User Manual
The ASL-DVS Suite

How to use the ASL-DVS web tools day to day: what every page, tab and button does, with a technical close to each chapter for when you want to know how it works. Written for the Pi Zero 2 W build; full-build differences are marked where they matter.

DASHBOARD SYSMON INSTMON WIFIMON 44HELPER

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
Dashboard	asl_dvs_dashboard_pi02w_v9_3_71_13_20261007.py	9.3.71.13-pi02w
SysMon	sysmon_pi02w_v6_13_67_9_20261007.py	6.13.67.9-pi02w
44helper	asl_dvs_m17_44helper_v0_0_175_20261008.py	0.0.175
instmon	instmon_v2_38_3_20261007.py	2.38.3
wifimon	wifimon_v5_28_20261007.py	5.28
Installer	install_asl_dvs_v6_6_20261006.sh	6.6
SysMon FULL	sysmon_v6_13_74_20261007.py	6.13.74
Dashboard FULL	asl_dvs_dashboard_v9_3_73_20261006.py	9.3.73

DRAFT · PART I OF V

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How to use this manual

If you are new to the suite, read Part I first, then turn to the chapter for the tool in front of you. Every chapter after Chapter 2 follows the same order, so you always know where to look:

1. **What it's for**: what the tool or tab does, and when you would open it.
2. **Before you start**: what has to be installed, and anything that can go wrong.
3. **Tab by tab**: each card and button, and what good and bad readings look like.
4. **Common tasks**: short recipes, step by step.
5. **Under the hood**: the technical close, in a purple-labelled panel. It covers the files, services, ports and timers involved. You can skip it on a first read.

Markers used in the text

NOTE Background, or something worth knowing.

TIP A quicker or safer way to do something.

CAREFUL This changes your node, restarts something or needs a reboot.

FULL BUILD Applies to the full build (Pi 3, 4 or 5) only. Also shown as the badge **FULL**.

Commands and pictures

Commands are typed on the Pi, in an SSH session or at a keyboard and screen. Lines starting with `#` are notes. Replace `<pi-address>` with your Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

Pictures marked **Illustration** are drawn to show a page's layout. They use the same tab names and order as the real page, but the values in them are made up. Example identities use the placeholders `N0CALL` and `<NODE>`.

PART I

Getting started

What the suite is, what each tool is for, how to log in, and the short version of installing and updating it. The full install steps are in the install manuals in the `docs/` folder.

1. The suite at a glance
2. Installing and updating, in brief

The suite at a glance

ASL-DVS is a set of five small web tools that run on the same Raspberry Pi as your AllStarLink node. Each tool has its own page in the browser. Together they let you run, check, repair and update the node without typing commands.

1.1 What is on the node

The tools sit on top of a working AllStarLink 3 (ASL3) node with DVSwitch. It helps to know the main parts, because the tools check and talk to all of them:

PART	WHAT IT DOES
Asterisk + app_rpt	The ASL3 engine. It runs your node, links to other AllStarLink nodes over IAX2 and handles EchoLink.
Analog_Bridge	Turns the node's audio into digital voice and back again. It talks to Asterisk on a private node (1999 by convention).
MMDVM_Bridge	Connects Analog_Bridge to the digital networks: DMR directly, and System Fusion, P25, NXDN and D-STAR through their gateways.
Gateways	YSFGateway, P25Gateway, NXDNGateway and ircDDBGateway. Each one connects to that mode's reflectors or talkgroups.
USRP2M17	Bridges a second private node (1917 by convention) to an M17 reflector.

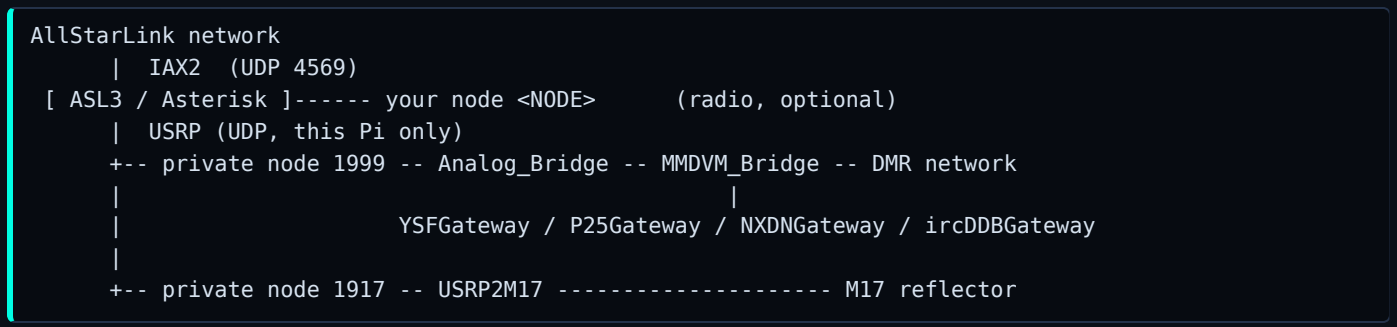


Figure 1.1 The audio path through a bridge node. Node numbers 1000 to 1999 are private: they never leave the Pi. The full layout is in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`.

1.2 The five tools

Each tool is one file, runs as its own service, and serves its own page on its own port.

TOOL	PORT	WHAT IT'S FOR	OPEN IT WHEN YOU WANT TO...
Dashboard	8989	Node control.	Connect to AllStar or EchoLink nodes, change DMR talkgroup or YSF, P25, NXDN, D-STAR, XLX or M17 reflector, use favorites, make a phone call.
SysMon	9999	System monitor and health checks.	See whether the Pi and node are healthy, find a stopped service or a bad config file, read logs, look after the SD card.
44helper	9997	Setup, repair and configuration helper.	Set up 44Net or Cloudflare remote access, install ASL3, DVSwitch, SVXLink, M17 or Ampersand, run updates, save and restore your config.

instmon	8990	Installer and manager for the suite.	Update the tools from GitHub, start or stop them, pause the suite for a big job, back up or image the SD card.
wifimon	8991	WiFi and power watchdog.	Change WiFi networks, check signal and supply voltage, set up the automatic safe shutdown.

> **NOTE** 44helper is optional. The other four make up the core suite. A sixth item, the Watchdog script, is no longer needed: the Dashboard's own service restarts it if it stops.

Which tool for which job

I WANT TO...	GO TO
Talk on a DMR talkgroup or a reflector	Dashboard, the mode's tab
Link to another AllStar node	Dashboard, ASL tab
Find out why something stopped working	SysMon, Overview and Services, then the matching check tab
Check that my config files agree with each other	SysMon, ASL-DVS and DVSM tabs; 44helper, Config tab
Save my config before I change it	44helper, Config tab (restore point)
Reach my node from outside my home network	44helper, Router Install / Tunnel tabs or Cloudflare
Update the suite	instmon, GitHub Updates
Update Linux and ASL packages	44helper, Update tab
Make the SD card last longer	SysMon, SD Card tab; 44helper, System Optimization
Change WiFi network	wifimon

1.3 A first look at each page

Every page has a header with the tool's name and version and a Log Out button. Most pages have a row of tabs under the header; click a tab to switch view. The pictures below show each tool's tabs in order.

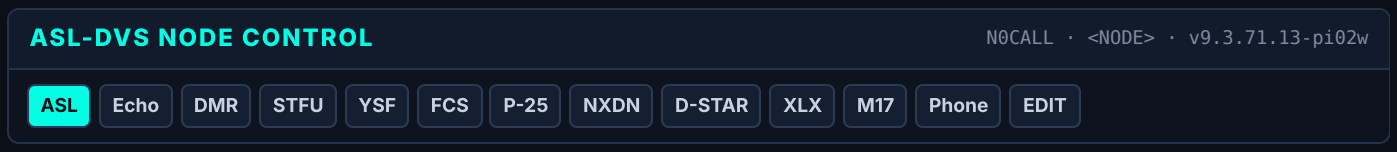


Figure 1.2 · Illustration. The Dashboard (port 8989). One tab per mode, plus Phone and EDIT. EDIT holds your favorites and settings. The Pi Zero 2 W build keeps every tab of the full build.

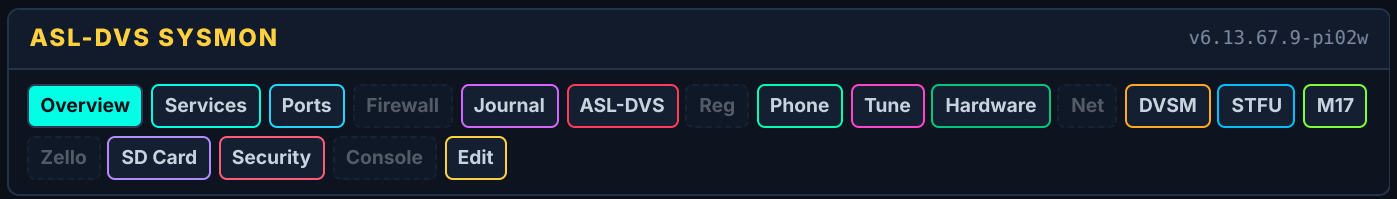


Figure 1.3 · Illustration. SysMon (port 9999). The 14 solid tabs are in the Pi Zero 2 W build. The 5 faded tabs (Firewall, Reg, Net, Zello, Console) are FULL build only.

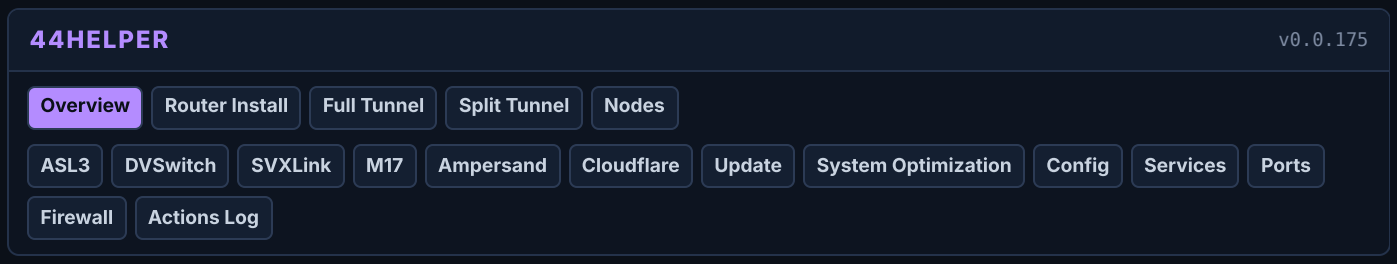


Figure 1.4 · Illustration. 44helper (port 9997). The first row is 44Net remote access. The second row has the installers, maintenance, Config and the status tabs. 44helper is the same on every Pi.

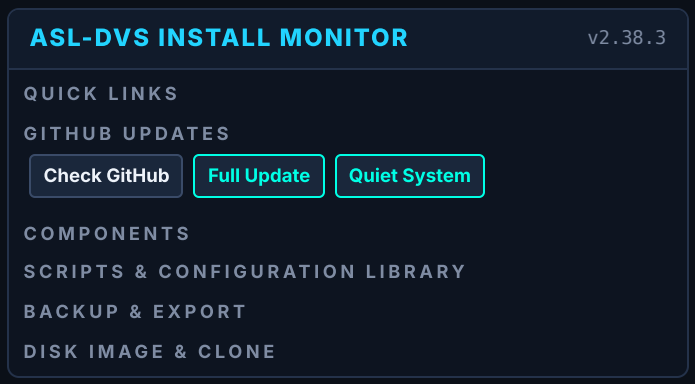


Figure 1.5 · Illustration. instmon (port 8990) is one long page of sections, not tabs.

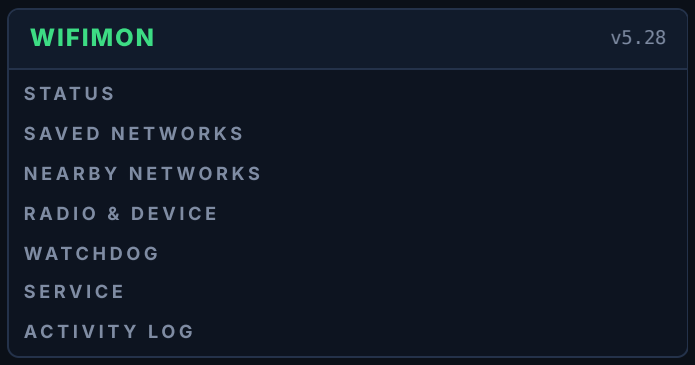


Figure 1.6 · Illustration. wifimon (port 8991) is also one page of cards.

1.4 Two builds: Pi Zero 2 W and full

The Dashboard and SysMon come in two builds. The other tools have a single build that runs on every Pi.

PI ZERO 2 W BUILD `PI02W`

For the 512 MB Pi Zero 2 W. This manual is written for it.

- Dashboard: every tab is kept. It searches the AllStar and EchoLink lists from disk instead of memory, and refreshes less often.
- SysMon: drops the Firewall, Reg, Net, Zello and Console tabs to save memory.

FULL BUILD `FULL`

For a Pi 3, 4 or 5.

- Dashboard: the same tabs, and it keeps the lists in memory.
- SysMon: all 19 tabs. Chapter 6 covers the five extra ones.

Which build am I running? Look at the version in the page header. A Pi Zero 2 W build version ends in `-pi02w`, for example `6.13.67.9-pi02w`. A full build version doesn't, for example `6.13.74`.

> **NOTE** Apart from SysMon, the builds look and work the same, so this manual points out the full build only where it differs.

1.5 Logging in

1. Open the page in a browser: `http://<pi-address>:<port>`, for example `http://192.168.1.50:9999` for SysMon.
2. Type the Pi's **root password** and press Log In. There is no user name.
3. When you're done, press **Log Out** in the header, especially on a shared computer.

- **The Dashboard and SysMon share one login.** Log in to either one and the other opens without asking, in the same browser. A shared login lasts 12 hours from the last time you used either page.
- **instmon, wifimon and 44helper each have their own login.**
- **Wrong passwords lock you out for a while.** On the Dashboard and SysMon, five wrong passwords within 5 minutes lock that address out of both for 15 minutes. The other tools have similar limits. Wait it out; don't keep guessing.
- **44helper stays logged in** until you log out, press **Log out all devices**, or the Pi reboots, because its jobs can run for a long time.

> **CAREFUL** The pages use plain HTTP, so your password crosses the network unencrypted. Only open them on your own network or over a VPN or tunnel, never straight from the internet. Chapters 7 and 8 cover safe remote access with 44helper.

> **TIP** If the login is always refused, set the root password again on the Pi with `sudo passwd root`. Raspberry Pi OS ships with root locked, so a new Pi needs this once.

1.6 Phones and tablets

The pages are laid out for phone, tablet and desktop widths. On a phone, wide tables hide their less important columns, and the tab row wraps onto more lines. Turning a tablet to landscape shows everything.

Leave a page open only while you're using it. Each open page polls its tool for fresh data, and on a Pi Zero 2 W that costs CPU and memory.

UNDER THE HOOD

Services, files and ports

TOOL	SYSTEMD SERVICE	PROGRAM	CONFIG
Dashboard	asl_dvs_dashboard	/usr/local/bin/asl_dvs_dashboard.py	/etc/asl_dvs/asl_dvs.conf
SysMon	sysmon	/usr/local/bin/sysmon.py	/etc/sysmon/sysmon.conf
44helper	44helper	/opt/44helper/asl_dvs_m17_44helper.py	/etc/44helper/44helper.conf
instmon	instmon	/usr/local/bin/instmon.py	/etc/asl_dvs/instmon_library/
wifimon	wifimon	/usr/local/bin/wifimon.py	/etc/wifimon/wifimon.conf

The Dashboard and SysMon program paths are links into `/usr/local/lib/asl_dvs/` and `/usr/local/lib/asl_dvs_sysmon/`. The installed file keeps its own name, so the version is always visible on disk.

One process each, started through the launcher

Every service starts its tool through the shared launcher `/usr/local/bin/asl_dvs_launch.py`:

```
ExecStart=/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py /usr/local/bin/sysmon.py
```

The launcher loads the tool as a module so that Python keeps its compiled copy in `__pycache__`. That roughly halves each tool's memory (SysMon on a Pi Zero 2 W: 54 MB started directly, 26 MB through the launcher). The launcher also turns on three memory savers: `python3 -OO`, `MALLOC_ARENA_MAX=2` and a periodic `malloc_trim`. Each has an off-switch file in `/etc/asl_dvs/` named `launch_no_optimize`, `launch_no_arena_cap` or `launch_no_trim`.

How a page works

- Each tool is a small Python HTTP server. The page's HTML, CSS and JavaScript are built into the tool's file, so there is nothing else to install and no web server to configure.
- The page fetches live data from the tool's own `/api/...` addresses as JSON and redraws its cards.
- The login checks the password against the root entry in `/etc/shadow`, and sets a session cookie. Requests that change anything also need a matching CSRF header, so another web page can't act for you.
- Version numbers come from the `VERSION` line in each file. The `-pi02w` suffix is how the installer and instmon recognise the Pi Zero 2 W build.

Checking on a tool from the shell

```
# is it running, and its last 30 log lines
systemctl status sysmon --no-pager
journalctl -u sysmon -n 30 --no-pager
# is anything listening on its port?
ss -tlnp | grep 9999
# restart all the web tools
sudo systemctl restart sysmon asl_dvs_dashboard instmon wifimon 44helper
```

Installing and updating, in brief

This chapter covers just enough to get the suite onto a node and keep it current. For every step and option, use the install manual for your build.

MANUAL	FILE IN DOCS/
Pi Zero 2 W install	ASL-DVS_Pi02w_Install_Manual.pdf
Full build install FULL	ASL-DVS_Full_Build_Install_Manual.pdf

2.1 Before you start

- A Raspberry Pi already running AllStarLink 3 with DVSwitch set up.
- SSH access with `sudo`, and a root password (`sudo passwd root`).
- On a Pi Zero 2 W, some free memory. Check with `free -m`. The install manual lists services you can turn off if you don't use them.

2.2 Install in three steps

1. **Get the files.** Download the current files from github.com/kd8pgk/ASL-DVS into one folder on the Pi.
2. **Install instmon first.** It can then install and update everything else from the browser.

```
sudo python3 instmon_v2_38_3_20261007.py --install
```

3. **Install the rest** with the installer. On a new node it asks for your callsign, node number and a label.

```
sudo bash install_asl_dvs_v6_6_20261006.sh
```

The installer picks the build for you: the Pi Zero 2 W build on a Pi Zero 2 W, the full build on anything else. If one build is already installed, it keeps that build. When it finishes, it checks that each tool is running and listening on its port.

> **TIP** 44helper is optional. Install it the same way, or from instmon's page, when you want its remote-access, installer or Config tools.

2.3 First visit to each page

PAGE	ADDRESS	CHECK FIRST
Dashboard	http://<pi-address>:8989	Your callsign and node number are right in the header.
SysMon	http://<pi-address>:9999	The Overview shows no red items (Chapter 3).
instmon	http://<pi-address>:8990	Every tool shows as installed and running.
wifimon	http://<pi-address>:8991	Answer "Turn the watchdogs on?" (below).
44helper	http://<pi-address>:9997	Take a restore point on the Config tab (Chapter 11).

wifimon's watchdogs

A new wifimon starts with both watchdogs off: it shows WiFi and voltage, but won't act on them. On your first login it asks whether to turn them on. **Add your WiFi networks first.**

- **WiFi watchdog:** reconnects to your saved networks, and shuts the Pi down after 3 minutes with no connection.
- **Voltage watchdog:** shuts the Pi down after 30 seconds of low supply voltage.

> **CAREFUL** Both watchdogs can shut the node down. A Pi Zero 2 W is sensitive to voltage drop, so use a good power supply and a short cable before turning the voltage watchdog on.

2.4 Keeping the suite up to date

Updates come from GitHub through instmon's **GitHub Updates** section.

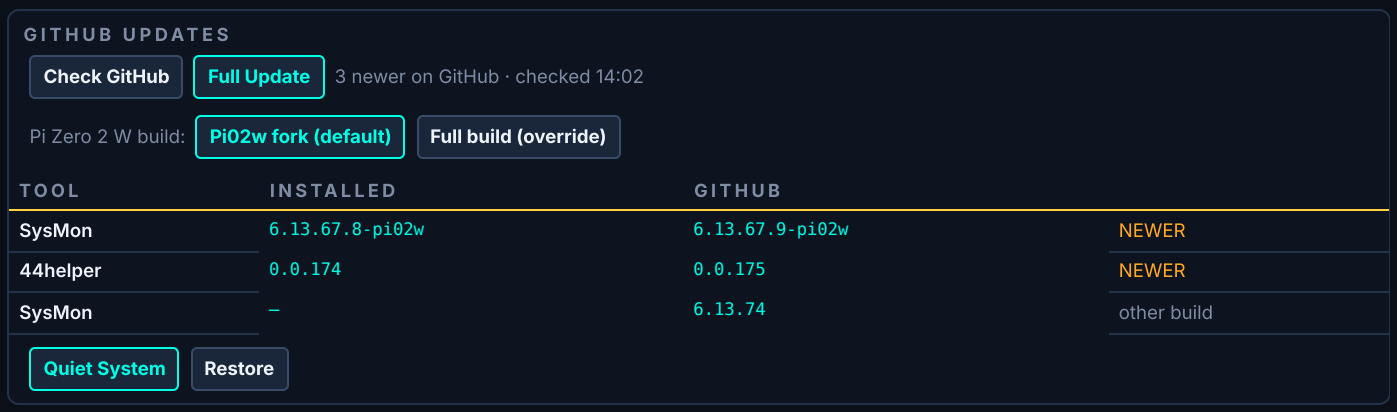


Figure 2.1 · Illustration. instmon's GitHub Updates section on a Pi Zero 2 W. The build choice appears only on a Pi Zero 2 W.

BUTTON	WHAT IT DOES
Check GitHub	Lists the newest copy of each tool on GitHub, and marks the ones newer than yours.
Update	Downloads and checks one file into instmon's library. Nothing is installed until you press Install on it.
Full Update	Updates every installed tool in one go, one tool at a time, with automatic roll-back if a new version fails.
Quiet System	Pauses SysMon, the Dashboard and 44helper during a big upload or install. The radio side keeps running, and everything comes back by itself after 30 minutes. **Restore** brings it back sooner.

> **TIP** Before a Full Update, take a restore point in 44helper's Config tab. Full Update only replaces the tools, but a restore point means you can roll back any config change you make while trying out the new versions.

Updating Linux, ASL3 and DVSwitch packages is a separate job. 44helper's Update tab handles it (Chapter 10).

2.5 Removing the suite

```
# remove one tool
sudo python3 /usr/local/bin/sysmon.py --uninstall
# remove every tool except instmon
sudo bash uninstall_asl_dvs_all_v1_3.sh
```

Your settings in `/etc/asl_dvs/`, instmon's library and wifimon's config stay in place. Remove instmon last, with its own `--uninstall`.

UNDER THE HOOD

What install and update actually do

--install, per tool

- Copies the file into place, writes its `.service` unit, and writes the shared launcher if it is missing.
- Enables the service so it starts at boot, and starts it.
- `--uninstall` stops and removes the service and the program. It removes the launcher only once no other service uses it.

How the installer picks a build

1. `SYSMON_VARIANT` / `DASH_VARIANT` set to `pi02w` or `full`, if you set them;
2. otherwise, the build already installed;
3. otherwise, `pi02w` on a Pi Zero 2 W and `full` on any other Pi.

If no file of the chosen build is found, that tool is skipped with a warning; the other build is never used in its place. The installer looks in its own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. `--non-interactive` (or `-y`) skips every question and takes the identity from `AUTO_CALLSIGN`, `AUTO_NODE` and `AUTO_LABEL`.

The checks on every download

Before instmon saves a file from GitHub, the file must match GitHub's file list. It must also have a version line and the right first line, pass a syntax check, and support `--install`.

Full Update, step by step

1. Downloads and checks every new file before anything changes.
2. Pauses the web tools.
3. For each tool in turn, saves a copy of the installed file in the library, runs the old `--uninstall`, and runs the new `--install`. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary. New install scripts go only to the Scripts library and are never run for you.

On a Pi Zero 2 W, the build choice decides which build Full Update installs. A tool installed in the other build is listed as **SWITCH** and replaced. On other Pis the choice is hidden and each tool keeps its build.

Settings that survive an update

Config files in `/etc` are never touched by an update. SysMon's Hardware tab options (HDMI and video driver off, lower GPU memory) are saved when SysMon is uninstalled during an update and put back on install, if the install follows within 30 minutes.

PART II

SysMon

SysMon is the node's health monitor. It shows whether every service is up, checks your Asterisk, DVSwitch and M17 config files against each other, reads logs, and looks after the hardware and SD card. Most of what it does is read-only. Every button that changes something says so.

3. SysMon basics and live status
4. Node and DVSwitch health
5. System care
6. Full-build tabs

SysMon basics and live status

This chapter covers the parts of SysMon you see every time: the header and command bar, then the four status tabs: Overview, Services, Ports and Journal. Open SysMon at `http://<pi-address>:9999`.

3.1 What SysMon is for

Open SysMon when something on the node isn't working, or when you want to be sure everything is. In one place it shows:

- whether Asterisk, the DVSwitch bridges, the gateways, USRP2M17 and the ASL-DVS tools are running;
- the Pi's load, memory, temperature and power health, updated every 5 seconds;
- which programs are listening on which network ports, and whether two of them clash;
- the system log (journal) of any service, filtered and ready to copy.

The status tabs can also start, stop and restart services. Everything else in SysMon is either read-only or asks before it changes a file.

3.2 The page at a glance

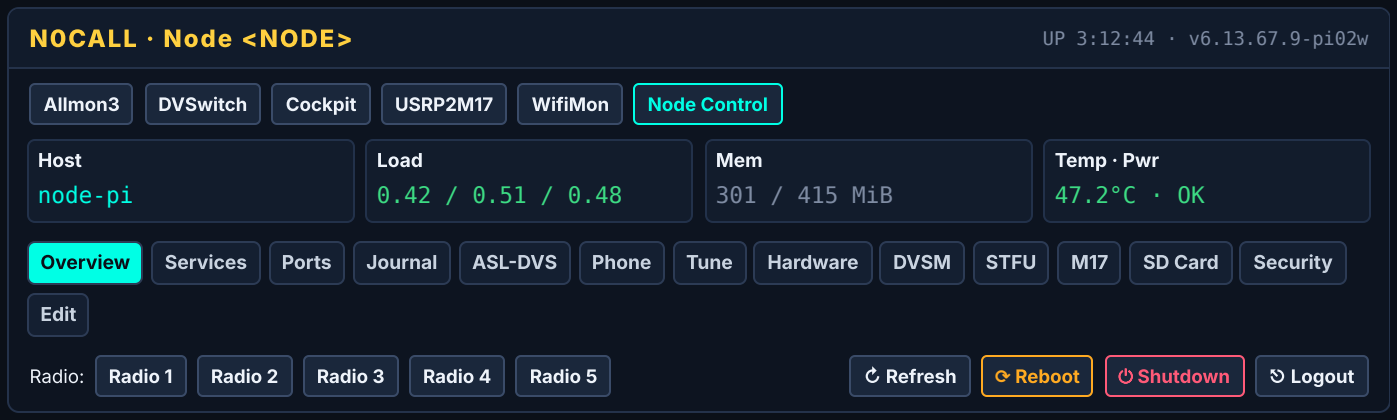


Figure 3.1 · Illustration. SysMon's top section. From the top: header, quick links, status strip, tabs and command bar.

PART	WHAT IT SHOWS OR DOES
Header	Your callsign and node number, how long the Pi has been up, and SysMon's version.
Quick links	Open Allmon3, the DVSwitch page, Cockpit, the USRP2M17 page and wifimon in a new tab. **Node Control** goes back to the Dashboard.
Status strip	Host name, load, memory, CPU temperature and power health (below).
Tabs	One per view. You can hide the ones you don't use on the Edit tab (Chapter 5).
Radio 1 to 5	Saved sets of radio audio settings. Pressing one applies it (Chapter 5, Tune tab).
Refresh	Reads the status strip again now, instead of waiting for the next 5-second update.
Reboot / Shutdown	Restart or power off the whole Pi. Both ask first. After Shutdown, the Pi stays off until you remove and restore power.

Reading the status strip

FIELD	GREEN	AMBER	RED
Load (1, 5, 15 min)	1-minute load up to 1.5	above 1.5	above 3
Temp	up to 60 °C	above 60 °C	above 75 °C
Pwr	`OK`	`WARN` : a problem happened since boot but isn't happening now	`⚡ UV` under-voltage now, or `⚡ THR` throttled now

Hover over **Pwr** to see exactly what was flagged, such as "Under-voltage since boot". When under-voltage starts, SysMon also shows a pop-up warning to check the power supply.

> **TIP** A Pi Zero 2 W has four 1 GHz cores and 512 MB, so a load of 1 to 2 during a tune or an update is normal. A load that *stays* red with nothing going on usually means a service is stuck restarting. Check the Overview for an `NR:` count.

The status dots

Every service list in SysMon uses the same coloured dots:

DOT	MEANS
● **Active**	Running normally.
● **Stopped**	Installed but not running. That is fine for a gateway you don't use, and a problem for Asterisk.
● **Failed / Masked**	It crashed or failed to start, or it has been blocked from starting (masked).
● **Unknown**	SysMon couldn't read its state.
● Not installed	There is no such service on this Pi. That is normal for optional parts.

Two banners you may see

- **"Restart sysmon to apply server address changes"**: you changed SysMon's port or bind address on the Edit tab. It takes effect when SysMon restarts.
- **"wifimon — Node Shutting Down"**: wifimon has started a safe shutdown because of low voltage or lost network. The Pi needs power-cycling, or its WiFi back, before it will come up again.

3.3 Overview tab

The Overview is SysMon's home page. It lists your **pinned services** in groups. Out of the box the groups are System, ASL-DVS, Asterisk Core and M17, which cover everything a bridge node runs.

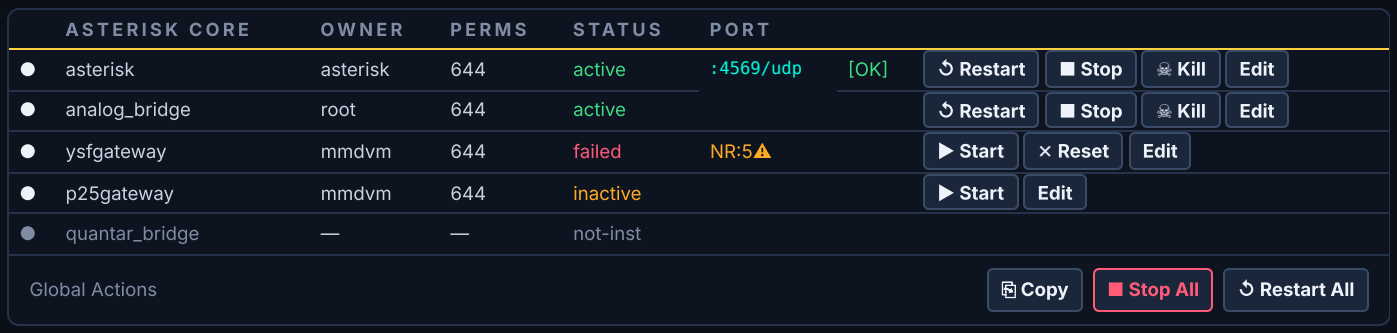


Figure 3.2 · Illustration. One Overview group. The buttons on each row depend on the service's state.

COLUMN	WHAT IT TELLS YOU
Name	The service. Click it to open its detail panel (3.4).
Owner	The Linux user it runs as. Amber means it runs as root but doesn't need to. Click the amber value for the Security tab's explanation.
Perms	The permissions on its service file. Amber or red means other users could change it. Click for the explanation.
Status	`active`, `inactive`, `failed`, `not-inst` or `unknown`.
Port	The port it should listen on. If it has restarted on its own since it was last started, this shows `NR:` and the count instead, with ⚠ above 3.
[OK] / [DUP]	For a running service with a port: `[DUP]` means another program has the same port and protocol, and hovering names both. `[OK]` means no clash.

Row buttons

STATE	BUTTONS
active	**Restart**; **Stop** (asks first); **Kill**, which ends the process at once with no clean shutdown (asks first); **Edit**, which opens the service file.
failed	**Start**; **Reset**, which clears the failed mark and the restart counter; **Edit**.
inactive	**Start**; **Edit**.

Restart is done carefully: SysMon stops the service, waits for it to stop, starts it, and waits for it to come up. The Overview refreshes itself a few times over the next half minute while that happens.

Global Actions

- **Copy** puts the whole Overview on the clipboard as text, ready to paste into a help request.
- **Stop All** stops every pinned service except SysMon itself.
- **Restart All** restarts every pinned service, SysMon included.

> **CAREFUL** **Stop All** and **Restart All** act at once, with no confirmation. Stop All takes the node off the air: it stops Asterisk, the bridges and the Dashboard. To get it back, press **Restart All** or reboot. SysMon keeps running, so the page stays usable.

3.4 Services tab

The Services tab lists **every** service on the Pi, not just the pinned ones.

- **Active** shows only running services. **All** also shows stopped, failed and disabled ones.
- Type in **filter by name** to narrow the list, for example `gateway`.
- The list refreshes every 30 seconds while the tab is open. **Copy** puts it on the clipboard as text.

The service detail panel

Click any service, here or on the Overview, to open its panel. The top shows badges for its state, whether it starts at boot, its process number, its restart count (`NR:`), CPU and memory use, and the user it runs as. Under that, **Ownership & permissions** shows who owns the service file and the program.

SECTION	BUTTON	WHAT IT DOES, IN PLAIN WORDS
Daemon Controls	Reload Daemon	Makes systemd re-read every service file. Needed after you edit one by hand.
	Reset Failed	Clears a "failed" mark so the service can be started cleanly.
	Mask / Unmask	Mask blocks a service from starting at all, even by hand, until you Unmask it. Asks first.
Runtime Controls	Start / Stop / Restart	As on the Overview. Stop asks first.
	Reload	Asks the program to re-read its settings without restarting. Greyed out for services that don't support it.
Boot Persistence	Enable / Disable	Whether the service starts by itself at boot. Disable asks first. Neither one starts or stops it now.
Pinned Services	Pin / Unpin	Adds the service to the Overview, in a group you pick or a new one, or removes it.
Status	(output)	The service's current status and its last 50 log lines, with errors in red and warnings in amber.

Editing a service file

Edit opens the service's systemd file in an editor, with **Save**, **Restart**, **Reload Daemon** and **Copy** buttons. After saving, press Reload Daemon and then Restart for the change to take effect.

> **NOTE** Service files that came with a package live in `/lib/systemd/system`. SysMon never changes those. Saving one creates your own copy in `/etc/systemd/system/`, which takes priority and survives package updates. A yellow bar warns you when this happens.

3.5 Ports tab

The Ports tab lists every network port a program on the Pi is listening on. It refreshes every 15 seconds.

COLUMN	MEANING
Dot	Green: the port belongs to a known service. Grey: a listener SysMon can't tie to a service.
Port / Proto	The port number, and TCP or UDP.
Process / PID	The program listening, and its process number.
Known Service	The service that program belongs to, or "— unknown —".
Conflict	`[OK]`, or `[DUP]` if more than one program has the port.

- **All / TCP / UDP** filter the list. **Copy table** copies what you see; **Copy all** copies every port.
- Click a row for its panel: the listen address, **Probe** buttons (HTTP and TCP for a TCP port, UDP for a UDP port) that test whether it answers, and for a known service **Restart Service**, **Stop Service** and **Journal**.

> **NOTE** `[OK]` only means nothing else is fighting for that port. It doesn't mean the service is healthy. A DVSwitch port clash is a common cause of "no audio". The pairing rules are in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`, section 4.

3.6 Journal tab

The Journal tab lists your pinned services that are installed, followed by any other running service. Click one to open its log in a pop-up window.

CONTROL	WHAT IT DOES
Lines 50 / 200 / All	How far back to read. All means the last 2,000 lines.
grep filter	Shows only lines containing your text, for example `error` or a talkgroup number.
Refresh	Reads the log again.
Restart / Start	Restart or start the service from the log window, so you can watch what happens.
Copy	Copies the lines you see (after the filter).

3.7 Common tasks

A service shows red

1. On the Overview, note the `NR:` count. A high count means it keeps crashing.
2. Open the Journal tab, click the service, choose 200 lines and filter on `error`.
3. Fix the cause. It is usually a config file, covered on the ASL-DVS, DVSM, STFU and M17 tabs (Chapter 4).
4. On the Overview, press **Reset**, then **Start**. Watch the dot turn green.

Find out what is using a port

1. Open the Ports tab and press UDP or TCP.
2. Find the port number. The Process and Known Service columns name the owner.
3. If it shows `[DUP]`, hover over it for both programs, then stop or reconfigure one of them.

Add a service to the Overview

1. Services tab, press **All**, and filter for the service.
2. Click it, pick a group under Pinned Services (or "New group...") and press **Pin**.

Send someone a status report

Use **Copy** on the Overview, the Services list, a Ports table or a journal window, and paste it into your message. Copies are plain text, so they paste cleanly into an email or forum post.

UNDER THE HOOD

How SysMon gathers and acts

Refresh timings

WHAT	HOW OFTEN	NOTES
Background poll (server)	every 10 s	Host, load, memory, swap, temperature, voltage, throttle state, disk, SD health, process count, wifimon shutdown state. Also tells systemd's watchdog that SysMon is alive.
Quick tests, port clashes	every 30 s	Every third background poll.
Status strip (page)	every 5 s	`GET /api/status`, read from the background poll's snapshot, so it costs almost nothing.
Services list	every 30 s	Only while the Services tab is open.
Ports list	every 15 s	Only while the Ports tab is open.
M17 reflector list	hourly check	Downloads `M17Hosts.json` again when it is stale (Chapter 4).

Switching tabs stops the previous tab's timer, so only the tab you're looking at polls.

Where the data comes from

- **Services:** `systemctl show` and `systemctl list-units`. The owner comes from the running process, or from `User=` in the unit.
- **Ports:** `ss -tlnpu`, with each process's command line from `/proc/<pid>/cmdline`.
- **Journal:** `journalctl -u <unit> -n <lines>`, between 10 and 2,000 lines.
- **Power:** `vcgencmd get_throttled`. Its bits give under-voltage, frequency cap, throttling and temperature limit, both now and since boot. Without it, SysMon shows the core voltage instead.

Service actions

Every action goes to `POST /api/svc`. Unit names are checked before anything runs.

- **Simple actions** (start, stop, reset-failed, enable, disable, reload, mask, unmask) run the matching `systemctl` command, with a 15-second limit.
- **Restart** runs in the background: `systemctl stop`, wait up to 15 s; if it is still running, `SIGKILL` and wait up to 5 s; then `systemctl start` and wait up to 10 s for it to become active. The result goes to SysMon's own log.
- **Kill** sends `SIGKILL` to the main process. systemd may then restart it, depending on the unit's `Restart=` setting.
- **Stop All** runs `systemctl stop` on each pinned unit except `sysmon.service`. **Restart All** runs `systemctl restart` on each pinned unit, SysMon included.

The pinned list

The Overview groups come from `[services] pinned` in `/etc/sysmon/sysmon.conf`, one line per entry:

```
__GROUP__:Asterisk Core
asterisk.service | ASL3 Asterisk/app_rpt | /etc/asterisk/rpt.conf | 4569 | udp
# unit | description | config_path | port | proto   ( - for none )
```

You can edit it as text on the Edit tab (Chapter 5). Pin and Unpin rewrite it for you.

Login and process

- SysMon runs as root, because it reads every config file and controls services. It refuses to start as any other user.
- SysMon and the Dashboard keep their logins in one shared file, `/run/asl_dvs/auth_session.json` (mode 600, locked while written). Both use the cookie `asl_dvs_session`, so one login covers both pages. A session expires 12 hours after it was last used. Five failures from one address within 5 minutes lock it out of both for 15 minutes. `/run` is cleared at reboot, so a reboot logs everyone out.
- The password is checked through PAM, with `/etc/shadow` as a fallback.
- The page is built once at start-up and sent compressed. Everything after that is small JSON requests.

```
# SysMon's own checks, without the web page
sudo python3 /usr/local/bin/sysmon.py --selftest
```

Node and DVSwitch health

Five SysMon tabs check the radio side of the node: ASL-DVS (the Asterisk, Allmon3 and DVSwitch config files), DVSM (DVSwitch Mobile accounts), STFU (BrandMeister), M17 (the USRP2M17 bridge) and Phone (autopatch). They read your files and tell you, in plain words, what is set, what is missing and how to fix it.

4.1 How the check tabs read

All five tabs use the same badges, so learn them once:

BADGE	MEANS	WHAT TO DO
PASS / OK	Set, and set to a real value rather than a default placeholder.	Nothing.
WARN	Works, but something is off or optional and missing.	Read the fix line; often it is fine to leave.
FAIL	Missing, still at a default such as `N0CALL`, or wrong.	Follow the fix line under it.
INFO	A value for your reference, such as a port or talkgroup.	Compare it with the other side of the link.
NONE	Listed but turned off (for example a module set to `noload`).	Nothing, unless you need it.

Many check lines end with a **fix**: the exact setting to add or the command to run. Copy buttons on each card give you the whole card as text.

> **CAREFUL** The **Edit** buttons on these tabs open the real config file, and **Save** overwrites it after one confirmation. SysMon writes the file safely (a power cut can't leave half a file), but it **does not keep a backup copy** of these files. Take a restore point in 44helper's Config tab first (Chapter 11), or press Copy before you change anything.

4.2 ASL-DVS tab

Four cards, one per group of files. Click a file row, or its **Edit** button, to open it in the editor.

/etc/asterisk/ — Config Files

Every `.conf` file in `/etc/asterisk`. The ones that matter most are checked under their name:

FILE	CHECKED
rpt.conf	Callsign is set and not a placeholder; each node number; its `rxchannel`, and for USRP nodes the RX and TX ports, with a reminder that Analog_Bridge must use the same pair.
manager.conf	At least one AMI user, and that each has a password.
echolink.conf	Callsign, password, EchoLink node number and the ASL node it links to.
rpt_http_registrations.conf	Each node that registers with AllStarLink, and that it has a password.
savenode.conf	Node and password pairs.
modules.conf	Which `chan_` channel drivers load, are required, or are set to `noload`. This list is folded away: press ▶ **Show**.

extensions.conf	The area code (`HOMENPA`, not the 999 default) and node number in `[globals]`.

Lines in a file that set a port get an `[OK]` or `[DUP]` badge, the same clash check as the Ports tab.

Allmon3 — Config Files

`allmon3.ini`, `web.ini`, `users`, `menu.ini` and `custom.css` in `/etc/allmon3`. `users` is read-only here: change Allmon3 logins with `allmon3-passwd`.

DVSwitch — Config Files

The DVSwitch files that exist on this Pi, from either their `/opt` or `/etc` location:

FILE	CHECKED
Analog_Bridge.ini	DMR ID and repeater ID (not defaults), AMBE mode, TX talkgroup and slot (1 or 2), colour code, software AMBE fallback, and the USRP RX and TX ports.
MMDVM_Bridge.ini	Callsign and DMR ID, location and description, and which modes (DMR, Fusion, P25, NXDN, D-STAR) are enabled, with their addresses. FAIL if no mode is enabled.
DVSwitch.ini	For each mode section: address and ports. For `[STFU]`, also the BrandMeister server, password, DMR ID and start talkgroup.
var.txt	The DMR networks the `dvs` menu knows about (BrandMeister, TGIF, DMR+, Other 1 and 2), and their addresses. Passwords are never shown.
Analog_Reflector.ini	If present: callsign, node ID and the USRP port.

D-Star — ircDDBGateway

Checks `/etc/ircddbgateway` for what the Dashboard's D-STAR gateway linking needs:

- the `ircddbgatewayd` service is running;
- `gatewayCallsign` is set, and `repeaterBand1` is a module letter. The card shows the 8-character name the Dashboard uses for the node, such as `N0CALL B`;
- `ircddbEnabled=1` (FAIL without it) and `dextraEnabled=1` (WARN);
- remote control (optional): when `remoteEnabled=1`, a password is set, the port is valid and something is listening on it;
- `gatewayAddress` is blank or one of this Pi's own addresses, never `127.x` and never the router's address.

Buttons: **Edit /etc/ircddbgateway**, **Restart ircddbgatewayd** (asks first; any D-STAR link drops), **Refresh** and **Copy**.

4.3 DVSM tab

DVSwitch Mobile (DVSM) is the phone app for talking through your node. This tab reads your files and shows exactly what to type into each of the app's four account types. The **Node IP** in the tab header is this Pi's address on your network.

CARD	USE IT FOR	KEY SETTINGS SHOWN
IAX2 Direct	Phone-mode connection straight to your node.	Host (LAN and WAN), port, user name and password from `iax.conf`, CallerID, node. "Phone Mode IAX2" ticked.

IAXRpt	**Recommended.** Radio mode: least data, and it only sends audio while the channel is active.	As above, but Caller Num must be `0` and "Phone Mode IAX2" unticked.
Node Mode	The app registers as its own AllStarLink node.	register.allstarlink.org, port 4569, and a **spare** node number registered to your callsign. Auto Connect unticked.
USRP / DVSwitch Bridge	Connect to Analog_Bridge's USRP port.	Host (local, LAN or WAN), the TX and RX ports, callsign and DMR ID.

Passwords show as `••••••••`. Click the eye to see one while you copy it into the app.

Compatibility Checks

SETTING (RPT.CONF NODE)	WHAT IT DOES	SHOWN AS
propagate_dtmf	Lets DTMF commands from DVSM reach the node.	INFO if off: only needed for DVSM.
phonesendlinks	Fills the DVSM Status tab's linked-node list.	INFO if off.
propagate_phonedtmf	Passes phone callers' tones to every linked node, bridges included.	**WARN if on.** Leave it off unless you need it.
remote_dtmf_allowed	Lets any linked node send commands to this node.	**WARN if on.**

It also checks `iax.conf`: that an `[iaxrpt]` (or callsign-named) user with a password exists, which IAXRpt needs, and how call tokens are set.

4.4 STFU tab

STFU is DVSwitch's BrandMeister terminal: it connects the node to BrandMeister directly, as a hotspot. Its settings live in the `[STFU]` section of `DVSwitch.ini`.

CARD	SHOWS
Installation	Whether the STFU program and its service are installed, running and enabled at boot.
Configuration	BrandMeister server, port (default 62031) and hotspot password (masked); your DMR ID; the USRP ports; start talkgroup; subscriber file; talker alias; log level. A link opens BrandMeister's master-server list.
Compatibility Checks	Server set, password set, DMR ID not a default, ports set, subscriber file found (it turns DMR IDs into callsigns), service running.
Sample [STFU] Stanza	A complete section built from your values, with placeholders for anything missing. **Copy Stanza** or **Edit Stanza**.
Edit DVSwitch.ini	Expand to edit the whole file. **Save**, then **Restart STFU** to apply.

> **TIP** The BrandMeister hotspot password is not your BrandMeister login password. Set it on brandmeister.network under **Self Care → Hotspot Security**.

4.5 M17 tab

The M17 bridge, USRP2M17, links private node 1917 to an M17 reflector. The Dashboard's M17 tab rewrites its settings file every time you connect or disconnect.

CARD	SHOWS
Installation	The program, its service and whether it is running. **Restart usrp2m17** asks first.
Configuration	`/opt/USRP2M17/USRP2M17.ini` : callsign; the reflector name and address (`0.0.0.0` means not connected); the M17 and USRP ports; gain; logging.
Compatibility Checks	Callsign set; reflector connected (WARN when idle, which is normal); USRP ports are 32008 and 34008; service running; `rpt.conf` has a `[1917]` node whose `rxchannel` ports match the ini file.
Reflector Host File	`/var/lib/mmdvm/M17Hosts.json` : how many reflectors it lists and how old it is. **Update Now** downloads a fresh copy.
Sample USRP2M17.ini	A complete file built from your values.
Edit USRP2M17.ini	The real file. Save, then Restart from the Installation card.

CAREFUL Edits to `USRP2M17.ini` don't survive the next M17 connect or disconnect on the Dashboard, which rewrites the whole file. Make lasting M17 changes on the Dashboard instead.

NOTE If `LocalPort` under `[M17 Network]` is missing, the sample fills in `32010` . That is the port MMDVM_Bridge uses for P25. The recommended value is `17010` (see the node configuration guide, section 4).

4.6 Phone tab

The Phone tab shows what Asterisk is set up to do for **autopatch** (radio dials out) and **reverse autopatch** (phone calls in to the radio). It is **read-only**. You set phone networks up on the Dashboard's Edit page, under Phone. The one exception is the Autopatch modules card.

CARD	WHAT IT TELLS YOU
Live	IDLE, DIALING, INCOMING or IN CALL; whether the phone node is linked to your node (it links while the Dashboard Phone tab is open); whether that tab is open (callers get busy when it isn't); which network is in use; phone patch on or off; calls now; sign-in state. Refreshes every 5 seconds.
Autopatch node	The private node the Dashboard made for each phone network.
Phone networks	Each network and its dialing target.
Hams Over IP	Shown only when you have a Hams Over IP account: the account and its sign-in.
Dialing rules	The dialplan contexts Asterisk has for each network.
Health checks	Asterisk running; nodes, files, live state and call routes all consistent. Each check is OK, WARN, FAIL or INFO, with a fix.

Router test	Three tests for calls that won't connect or have one-way audio (below).
Autopatch modules	The Asterisk modules each feature needs, and whether each is loaded. You can change them here (below).
Recent call attempts	Up to 20 phone calls from the last 24 hours, in or out, with the network and the last 4 digits only.

Router test

1. **Going out: Run outgoing test** signs in to your phone provider once and sends one STUN packet. It takes about 5 seconds.
2. **Call audio:** start a **Test call** on the Dashboard's Phone tab, then press **Check audio now**. SysMon counts audio packets in and out over 4 seconds. "Out but none in" points to the router forward or the Pi firewall. Audio in but none out is normal on a simplex node while nobody talks.
3. **Coming in:** press **Start watching (10 min)**, have another Hams Over IP member call your extension, then press **Check**.

Autopatch modules

Modules are grouped by feature: **Autopatch** and **Reverse autopatch**, each READY, MISSING, OFF, NOT SET UP or ASTERISK DOWN. Modules both features need, and the ones every node needs, are locked (🔒).

- **Activate / Deactivate** a feature: sets every module it needs to load and starts it, or sets the modules only it uses to noload and stops them. The confirmation lists exactly which modules.
- Per module: **Load** / **No load** change `modules.conf` for the next Asterisk restart; **Start** / **Stop** load or unload it now.
- If a module file is missing from the Pi, the card names the package it comes from and offers **Reinstall**, with the same commands shown for doing it by hand.

> **NOTE** Unlike the file editors, this card backs up `modules.conf` before every change. It keeps the last 5 copies next to the file.

The full story of the phone dialplan is in `docs/ASL-DVS_Pi02w_Phone_Dialplan_Guide.pdf`.

4.7 Common tasks

No audio between AllStar and DMR

1. ASL-DVS tab: under `rpt.conf`, note node 1999's RX and TX ports.
2. Under `Analog_Bridge.ini`, check that its RX Port and TX Port are the same pair the other way round, as the note says.
3. Ports tab: check that neither port shows `[DUP]`.
4. Fix the file with Edit, then restart `analog_bridge` and `asterisk` on the Overview.

Set up DVSwitch Mobile

1. DVSM tab: make sure the IAXRpt card shows PASS. If not, follow the Compatibility Checks fix.
2. In the app, add an IAX2 account and copy each field from the IAXRpt card. Use Host (LAN) at home and Host (WAN) away.

3. Untick Phone Mode IAX2 and set Caller Num to 0.

Get a new node ready for STFU

1. STFU tab: work down the Compatibility Checks until every line shows PASS.
2. Copy the Sample [STFU] Stanza if your file has no `[STFU]` section, and paste it in with Edit DVSwitch.ini.
3. Save, then Restart STFU.

UNDER THE HOOD

Files, endpoints and what is sent to the page

Where each tab reads

TAB	FILES AND COMMANDS	ENDPOINT
ASL-DVS	every `*.conf` in `/etc/asterisk`; `/etc/allmon3/*`; the DVSwitch files under `/opt`, `/etc` and `/var/lib/dvswitch/dvs/var.txt`; `/etc/ircddbgateway`, plus `ss -Hlunp` and `ip -o -4 addr`	`/api/asterisk/files`, `/api/allmon3/files`, `/api/dvswitch/files`, `/api/dstar-gw`
DVSM	`iax.conf`, `rpt.conf`, Analog_Bridge's ini; the LAN address from the default route	`/api/dvsm`
STFU	`DVSwitch.ini [STFU]`, `/opt/STFU`, `stfu.service`	`/api/stfu`
M17	`/opt/USRP2M17/USRP2M17.ini`, `rpt.conf [1917]`, `usrp2m17.service`, `M17Hosts.json`	`/api/m17`
Phone	Asterisk's files and the CLI (`asterisk -rx`), `/var/log/asterisk/full` or `messages` (last 2 MB), or the journal	`/api/phone`, `/api/phone/routertest`, `/api/phone/modules`

Outside connections

- **DVSM** finds your public (WAN) address from `api.ipify.org`, falling back to `checkip.amazonaws.com`, with a 2-second limit each. Without internet the WAN rows say "Could not detect".
- **M17 host file** comes from `hostfiles.refcheck.radio`, falling back to `m17-project.github.io`. A download is only kept if it is at least 2 KB and lists at least 10 reflectors. The background poll checks every hour and fetches a new copy when the file is over 24 hours old.

Secrets

- The DVSM and STFU cards send the account passwords to your logged-in page, masked until you press the eye, so you can copy them into an app.
- The ASL-DVS checks report only whether a password is set, never its value. `var.txt` password lines are skipped completely.

Writes

- File saves go to `POST /api/asterisk/file`, `/api/dvswitch/file` and `/api/allmon3/file`. Each writes a temporary file, keeps the old file's owner and mode, syncs it to disk, and swaps it in place. No backup is kept. Asterisk file names are validated, and the DVSwitch and Allmon3 editors only accept the files in their fixed lists.
- Autopatch module changes edit only SysMon's own marked block in `modules.conf`, between `; >>> sysmon-phone modules` and `; <<< sysmon-phone <<<`. A conflicting `noload` elsewhere is commented out with `; sysmon-phone off:`. If the Dashboard's own block says noload, SysMon warns instead and leaves it, because Save Phone on the Dashboard would undo the change. Backups are named `modules.conf.sysmon-bak-YYYYMMDD-HHMMSS`, keeping 5. Live start and stop use `module load` / `module unload` through the Asterisk CLI.
- Restarts use `systemctl restart` on `stfu`, `usrp2m17` or `ircddbgatewayd`.

Defaults the checks treat as "not set"

Callsigns `N0CALL`, `NOTSET`, `NONE` or blank; the stock placeholder DMR and repeater IDs (such as `123456789`); the stock STFU server; area code `999`; and M17 address `0.0.0.0`.